

The AI studio presents

Computational Media

What did you just say?

*Taking the pulse of AI models
Spring 2026*

Course Information

Instructor: Professor Marc Böhlen (mrbohlen@buffalo.edu)
ART 384 / ART 543 - Registration#: 23052, 3 credits
Location: CFA136
Times: Monday and Wednesday, 15:30 – 18:00
Office hours: Monday, 15:00
3 credits

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Course Description

This advanced undergraduate and graduate course introduces students to the art and science of probing the behavior of Generative Artificial Intelligence (GenAI) systems through critical thinking and AI literacy. Artificial Intelligence—particularly its generative branch GenAI—has become a site of cultural and technical transformation, reshaping how knowledge is produced and interpreted across disciplines. Unlike traditional software with deterministic outputs, GenAI systems can produce varied and context-dependent responses to the same input, making the inquiry of these systems both technically and critically complex. The course begins with a survey of early generative paradigms such as genetic algorithms before turning to the foundational mechanisms of modern Large Language Models. Core topics include transformer architecture, reinforcement learning, and reward modeling. Building on these, we will examine downstream issues such as alignment, jailbreaking (techniques to bypass model constraints), and red teaming (methods for identifying vulnerabilities).

Students will apply these concepts in guided, hands-on explorations of contemporary AI systems developed by frontier labs (e.g., Google, Meta, OpenAI, Anthropic). Following introductory exercises, student teams will design mixed methods experiments to document and analyze model behavior in context. Exercises will unfold in open-source tools and environments such as Google CoLab, computational notebooks, and remote scripting frameworks. Students will learn how to construct multi-shot prompts, programmatically interface with AI models, and assess system outputs using text-based metrics such as relevance, coherence, and specificity, as well as

correlation analyses for bias detection. Alongside these technical investigations, the course integrates critical and humanities-based frameworks - including concepts such as techno slop, kitsch, and nostalgia - to sharpen the interpretive analysis of GenAI artifacts.

Spanning all these activities we will seek to understand how the evolving landscape of AI systems is impacting trust, authorship, and knowledge production across disciplines.

Deliverables

- a) 3 exercises
- b) 1 group project
- c) 1 semester project

Learning Outcomes

This course is designed for ambitious students. It requires above average commitment and determination. Expect to invest beyond class time at least 1.5 hours per day, 5 days a week to get an A grade. The main learning outcomes each student should achieve are:

- 1) Gain basic literacy of how generative AI systems produce novel output
- 2) Understand the challenges of probing generative AI systems.
- 3) Develop a nuanced approach to the use of generative AI systems in context.

Assessment

- 1) This knowledge will be evaluated in assignments # a, b
- 2) This ability will be evaluated in # b, c

Grading Policy

a - 60%, b - 20%, c - 20% (total 100)

Grading Scale

100-90: A; 89-80: B; 79-70: C; 69-60: D (less than 60: F)

UB Curriculum Capstone

If you are completing this course as part of your UB Curriculum requirements, please select an 'artifact' from this course that is representative of your learning and upload it to your ePortfolio account. Typically, the final project will serve as such an artifact. However, you can choose an alternate submission, including homework assignments, exams, research papers, projects, lab reports, presentations. Your final UB Curriculum requirement, UBC 399: UB Curriculum Capstone, will require you to submit these 'artifacts' as you process and reflect on your achievement and growth through the UB Curriculum. For more information, see the UB Curriculum Capstone website: <https://www.buffalo.edu/ubcurriculum/capstone.html>.

Syllabus (Details subject to change)

Week	Topic	Assignments
W1	Course overview	
W2	A simple neural network (with all the details)	
W3	Vectorization and latent space	
W4	Stable diffusion	
W5	Reinforcement learning	A1
W6	Transformer architectures	
W7	Prompt design	
W8	System prompt design	A2
W9	SPRING BREAK	
W10	Survey of research labs studying GenAI behavior: Haize, METR, Apollo Research	
W11	Multi-prompt design	A3
W12	Iterative prompting and role playing	
W13	Metrics of quality assessment (BLEU, consistency, coherence)	
W14	Bias audits (filtering with deny lists)	A4 (group)
W15	Edge case testing	
W16	Jailbreaking and red teaming	
		Semester project

Relevant UB Policies

Academic Integrity

Academic integrity is a fundamental university value. Through the honest completion of academic work, students sustain the integrity of the university while facilitating the university's imperative for the transmission of knowledge and culture based upon the generation of new and innovative ideas. Students are responsible for the independent and honest completion and representation of their work, for the appropriate citation of sources, and respect for other's academic endeavors. By placing their name on academic work, students certify the originality of all work not otherwise identified by appropriate acknowledgements. Details at <http://www.student-affairs.buffalo.edu/judicial/swj.php>

Contract cheating will be managed as plagiarism. Contract cheating is a form of academic dishonesty in which students have academic work completed on their behalf for payment. Some examples of contract cheating include the use of Chegg, Course Hero, or other websites to receive answers to questions.

Accessibility Resources

If you have any disability which requires reasonable accommodation to enable you to participate in this course, please contact the Office of Accessibility Resources, 25 Capen Hall, 645-2608, and the instructor of this course. The Office of [Accessibility Resources](#) will provide you with information and review appropriate arrangements for reasonable accommodations.

Mental Well-Being

As a student you may experience a range of challenges that can cause barriers to learning or reduce your ability to participate in daily activities. These might include strained relationships, anxiety, elevated levels of stress, alcohol/drug problems, feeling down, health concerns, or unwanted sexual experiences. Counseling, Health Services, and Health Promotion are here to help with these or other issues you may experience. You can learn more about these programs and services by contacting:

Counseling Services:

120 Richmond Quad (North Campus), 716-645-2720

202 Michael Hall (South Campus), phone: 716-829-5800

Health Services: Michael Hall (South Campus), 716- 829-3316

Health Promotion: 114 Student Union (North Campus), 716- 645-2837